

5.1 Brief tutorial on PIC

We will be using a PIC microcontroller for our project because of its versatility. So it will be good if we learn about PIC microcontrollers before we start working on them. This chapter provides basic knowledge which you would need to know before starting on with PICs.

PIC microcontrollers are a product from Microchip technology Inc. It's not clearly known whether PIC is an abbreviated form of something or not. Some people say it stands for Peripheral Interface Controller, and some say it stands for Programmable Interface Controller. The PIC microcontrollers are favourites of many hobbyists and professionals, because of following reasons:

1.Availability: PIC microcontrollers are very easily available in every electronics markets as well as online. One of the important reasons why they were a hit with hobbyists was that earlier Microchip used to provide free samples of PIC microcontrollers. So people who were starting with microcontrollers used to order them and once they learnt using it, they used to stick with it. Sadly, now Microchip asks you to pay shipping charges to India, which is higher as compared to the cost of PICs you can buy from the Indian markets.

2.Extensive Product Range: The PICs are available in numerous models and with different features, to suit the requirement of even the most demanding circuit designer.

3.Support: Since a lot of people use PICs, so a lot of online guides, forums, webpages and books are available on projects based on PICs. Also, the datasheets provided by Microchip for each PIC is very good and exhaustive.

4.Price: The PIC microcontrollers are relatively cheap and a basic PIC like PIC16F84, which would suit our application, would cost around ₹100.

A lot of people say that the architecture of certain PICs is tedious and is not helpful while programming in assembly language, but we need not worry about it since we will be programming in C++ language.

5.1.1 PIC: Internal structure

A PIC is a complete computer in itself. In the figure given alongside, a PIC16F84 microcontroller is shown. Any PIC would have three basic parts:

- 1.Central Processing Unit (CPU)
- 2.Program Memory (PROM)
- 3.Input Output Ports (I/O Ports)

The CPU is the 'brain' of the microcontroller. It reads the instructions